Pathfinders

PathWise AI
Project Plan

Version 1.0
02 24 2026

Release Checklist

Checklist Item	Control (Yes/No)
The document has passed the quality assurance process.	YES
The document is visually appealing and has a professional look and structure.	YES
All placeholders such as text within <> are replaced with appropriate text.	YES
The document is prepared in the “Justify Text” format.	YES
The document does not include typos, different fonts, and misalignments.	YES
In the revision history, the first version states “The first version”. Additional versions include an entry highlighting all the changes from the previous version.	YES
The table of contents (TOC) is updated as necessary.	YES
References list other project documents and additional standards or documents needed to understand the document. The references include the necessary document/artifact version number and publication date.	YES
The language grammar is used appropriately.	YES
Sentences are written as full sentences.	YES
The wording in the document is formal.	YES
The wording in the document is clear.	YES
The wording in the document is professional.	YES
All project-related definitions, acronyms, and abbreviations sections are included in the related tables.	YES
The team member names are specifically spelled out in the document preparation and approvals sections.	YES
The purpose and scope of the document are adequately described.	YES
All sections include the necessary and adequate information.	YES
The business goals are about how to secure the team/company's place in the market. The product goals are about all the goals of the product. The quality goals list all the product quality attributes for the product.	YES
The deliverables list all the deliverables including all project artifacts (document, code, presentation, etc).	YES
The project risks appropriately and in full sentences.	YES
The work package table is completed with all necessary fields.	YES

Final Control	Yes/No
The document is checked based on the “Team Project Artifacts Common Errors” Guideline. The document is ready for release.	YES
The document is not ready for release.	NO

Document Preparation

Name	Role	Approval (Signature)	Approval Date
Meghana Nalluri	Project Requirements Manager	MN	02/24/26
Bharad Jakkula	Project Requirements Engineer	BJ	02/24/26
Sricharitha Katta	Project Configuration Manager	SK	02/24/26
Vineeth Reddy Kodakandla	Project Manager	VK	02/24/26

Document Approvals

Name	Role	Approval (Signature)	Approval Date
Bharadwaj Jakkula	Project Design Manager	BJ	02/24/26
Sricharitha Katta	Project Configuration Manager	SK	02/24/26
Bharadwaj Jakkula	Project Test Manager	BJ	02/24/26
Vineeth Reddy Kodakandla	Project Risk Manager	VK	02/24/26
Sricharitha Katta	Project Technical Manager	SK	02/24/26
Vineeth Reddy Kodakandla	Project Manager	VK	02/24/26
NA	Project Sponsor	NA	NA

Revision History

Date	Version	Description	Author
02/24/26	1.0	The First Version.	Bharadwaj Jakkula

Table of Contents

1. Introduction 6
   - 1.1 Purpose of the Document 6
   - 1.2 Scope of the Document 6
   - 1.3 References 6
   - 1.4 Definitions, Acronyms, and Abbreviations 6
2. Project Goals 7
   - 2.1 Business Goals 7
   - 2.2 Product Goals 7
   - 2.3 Quality Goals 7
3. Project Stakeholders and Stakes 7
   - 3.1 Summary of Key Stakeholder or User Needs 8
4. Project Budget 8
5. Project Milestones and Schedule 9
6. Project Communications 9
7. Project Members and Roles 10
8. Project Work Breakdown Structure (WBS) 11
9. Project Deliverables 12
10. Project Risks 13

Project Plan

1. Introduction

1.1 Purpose of the Document

The purpose of this document is to lay out the comprehensive plan for the PathWise AI project. This document defines the project goals, stakeholder needs, budget, milestones, team roles, work breakdown structure, deliverables, and risks associated with the development of the PathWise AI system. It serves as the primary planning reference for all team members and stakeholders throughout the project lifecycle.

1.2 Scope of the Document

This document provides a complete overview of the project planning for PathWise AI, an AI-powered SD-WAN management platform. It encompasses the business, product, and quality goals that guide development activities, the identification of key stakeholders and their needs, the project budget and resource allocation, the milestone schedule aligned with the academic semester, team member roles and responsibilities, a detailed work breakdown structure, the complete list of project deliverables, and a risk assessment with mitigation strategies. This document is intended for all project stakeholders including the development team, the course instructor, and potential end users.

1.3 References

The following documents are referenced within this project plan:

1. Team Pathfinders. PathWise AI – Team Project Proposal, Version 1.0, February 2026.
2. Team Pathfinders. PathWise AI – Project Vision Document (PVD), Version 1.3, February 2026.
3. Hochreiter, S., & Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
4. Open Networking Foundation (ONF). Software-Defined Networking (SDN) Definition. ONF White Paper.
5. OpenDaylight Project. OpenDaylight SDN Controller Documentation. https://www.opendaylight.org.
6. Lantz, B., Heller, B., & McKeown, N. (2010). A Network in a Laptop: Rapid Prototyping for Software-Defined Networks. ACM SIGCOMM.
7. Batfish Team. Batfish: A Network Configuration Analysis Tool. https://batfish.org.
8. Pressman, R. S. Software Engineering: A Practitioner's Approach. McGraw-Hill Education.

1.4 Definitions, Acronyms, and Abbreviations

Term	Abbreviation / Acronym	Definition
Software-Defined Wide Area Network	SD-WAN	A virtual WAN architecture that allows enterprises to leverage any combination of transport services to securely connect users to applications.
Software-Defined Networking	SDN	An approach to networking that uses software-based controllers or APIs to direct traffic on the network.
Long Short-Term Memory	LSTM	A type of recurrent neural network architecture designed to learn long-term dependencies in sequential data.
Intent-Based Networking	IBN	A form of network administration that incorporates machine learning and orchestration to automate administrative tasks.
Small-to-Medium Enterprise	SME	Organizations with fewer than 500 employees and revenue typically under $50 million.
Application Programming Interface	API	A set of protocols and tools for building software applications.
Digital Twin	DT	A virtual representation of a physical system used for simulation and testing.
Command Line Interface	CLI	A text-based interface used to interact with software and operating systems.

Software Requirements Specification	SRS	A document that describes what the software will do and how it will be expected to perform.
Software Design Document	SDD	A document that describes the architecture and design of a software system.
Software Test Document	STD	A document that describes the test plan, test cases, and test results for a software system.
Work Breakdown Structure	WBS	A hierarchical decomposition of the total scope of work to be carried out by the project team.
Voice over Internet Protocol	VoIP	A technology that allows voice calls to be made over the internet.
Project Plan	PP	A formal document that defines how the project is executed, monitored, and controlled.

2. Project Goals

2.1 Business Goals

The primary business goal of PathWise AI is to establish a competitive position in the enterprise networking market by addressing the critical gap between expensive proprietary SD-WAN solutions and the needs of Small-to-Medium Enterprises (SMEs). The platform aims to democratize access to enterprise-grade network reliability by offering a vendor-agnostic, AI-powered solution at a fraction of the cost of proprietary alternatives such as Cisco Viptela, VMware VeloCloud, and Fortinet Secure SD-WAN. By targeting SMEs, Managed Service Providers (MSPs), healthcare facilities, educational institutions, and retail chains, PathWise AI seeks to capture a significant share of the underserved market segment that requires predictive network intelligence without prohibitive capital expenditure or specialized staffing requirements.

2.2 Product Goals

The product goals for PathWise AI are organized around five core capabilities. The first goal is to develop and train a Long Short-Term Memory (LSTM) neural network capable of predicting network link health, including latency, jitter, and packet loss, for a future window of 30–60 seconds with an accuracy of at least 90% as measured by Mean Squared Error. The second goal is to implement a closed-loop Software-Defined Networking (SDN) architecture where the control plane autonomously re-routes mission-critical traffic based on AI predictions, achieving hitless handoff with zero packet loss. The third goal is to build a Digital Twin Validation Sandbox using Mininet and Batfish that acts as a safety validation layer for every AI-generated routing change before production deployment. The fourth goal is to create an Intent-Based Networking (IBN) interface that allows generalist IT staff to manage complex network policies using natural language commands without requiring specialized CLI certifications. The fifth goal is to provide a Multi-Link Health Scoreboard that offers real-time visualization of link health scores, AI prediction confidence levels, and transparent reasoning behind every automated path switch.

2.3 Quality Goals

The quality goals for PathWise AI define the measurable quality attributes that the system must achieve. Prediction accuracy must be at least 90% for 30–60 second latency, jitter, and packet loss predictions. System availability must be at least 99.9% to ensure continuous network optimization. Failover time must be less than 50 milliseconds to achieve hitless handoff without packet loss. Digital Twin sandbox validation must complete within 5 seconds to enable timely production deployment. The Intent-Based Networking dashboard must respond within 2 seconds for all operations and natural language query processing. The system must scale to support management of at least 100 network sites with multiple WAN links each. All common administrative tasks must be accomplishable through the IBN interface without command-line knowledge.

3. Project Stakeholders and Stakes

To effectively provide products and services that meet stakeholders' and users' real needs, it is necessary to identify and involve all stakeholders as part of the requirements modeling process. The stakeholders for PathWise AI include the Course Instructor who evaluates project deliverables and ensures sound software engineering practices, SME Business Owners and IT Decision Makers who would fund and adopt the platform, Network Administrators and IT

Teams who manage day-to-day network operations, Managed Service Providers who deploy the system across multiple client sites, End Beneficiaries who rely on stable network connectivity, and Network Security and Compliance Officers who validate that AI-driven changes comply with security policies and regulations.

3.1 Summary of Key Stakeholder or User Needs

The following key problems with existing solutions have been identified through stakeholder analysis.

The first and most critical need is addressing reactive network failure detection that causes downtime and dropped packets. Traditional SD-WAN systems rely on threshold-based monitoring that only triggers failover after degradation has already occurred, creating the "switching gap" where packets are lost during the transition. Stakeholders require a predictive system using LSTM-based deep learning that forecasts link brownouts 30–60 seconds in advance and executes hitless traffic steering with zero packet loss before degradation impacts users.

The second critical need is eliminating the complex CLI-based SD-WAN management that requires specialized expertise. Configuring SD-WAN policies requires knowledge of CLI commands, YANG/NETCONF, and vendor-specific syntax, which SMEs often lack. Stakeholders require an Intent-Based Networking interface that allows generalist IT staff to manage complex network policies using natural language commands.

The third critical need is ensuring that AI-driven routing changes do not risk destabilizing the production network. Autonomous AI decisions are perceived as black box operations, and without validation, incorrect routing changes could introduce loops, violate firewall policies, or crash the live network. Stakeholders require a Digital Twin Validation Sandbox that simulates every AI-driven routing change in a virtual environment before production deployment.

The fourth need is providing transparency in AI decision-making to build trust. Network administrators and business leaders need to understand why the AI made a specific routing decision. Stakeholders require a Multi-Link Health Scoreboard providing real-time visualization of link health scores, AI prediction confidence levels, and clear reasoning behind every automated path switch.

The fifth need is making enterprise-grade SD-WAN solutions affordable for SMEs. Proprietary hardware and licensing from vendors like Cisco Viptela or VMware VeloCloud require significant capital expenditure. Stakeholders require a vendor-agnostic, software-based solution that delivers enterprise-grade predictive reliability on commodity hardware.

The sixth need is simplifying the management of multiple WAN links across distributed locations. Organizations with multiple branches use diverse WAN links such as Fiber, Satellite, and 5G, each with different performance characteristics. Stakeholders require unified multi-link health monitoring with autonomous, predictive traffic steering managed from a single centralized dashboard.

4. Project Budget

Budget Component	Estimated Effort (Man*Hours)	Percentage
Project Management and Planning	80	13%
Requirements Analysis and Documentation	80	13%
System Architecture and Design	90	15%
AI/ML Model Development (LSTM Engine)	100	17%
Backend and SDN Integration Development	80	13%
Frontend Dashboard Development	60	10%

Digital Twin Sandbox Development	40	7%
Testing and Quality Assurance	40	7%
Configuration Management and DevOps	20	3%
Project Presentations	10	2%
Total	**600**	**100%**

5. Project Milestones and Schedule

The following table presents the project milestones, scheduled dates, and associated deliverables for the PathWise AI project.

Project Start Date: 02/03/2026

Project End Date: 05/08/2026

Project Milestone	Date	Deliverable
Problem Analysis, Vision Creation, Project Planning	02/17/2026	Product Vision Document Project Plan
Requirements Analysis	03/09/2026	Software Requirements Specification (SRS)
Product Design	03/30/2026	Software Design Document (SDD)
Product Implementation	04/14/2026	Software Code
Product Test	04/28/2026	Software Test Document (STD)
Final Product Deliverable	05/01/2026	All Documentation and Code
Product Launch	05/08/2026	Project Presentation

6. Project Communications

The following communication plan establishes the protocols for information exchange among team members and stakeholders throughout the PathWise AI project lifecycle.

Project Communication Type	Responsible Team Member	Receiving Stakeholder	Frequency	Medium	Scheduled Time
Daily Project Communication	Any Team Member	Any Team Member	Anytime	E-mail, etc.	Anytime
Team Meetings	Vineeth Reddy Kodakandla (Project Manager)	All Team Members	Weekly	Face-to-face / Online Zoom	Every Monday and Thursday.

Project Progress Reports	Vineeth Reddy Kodakandla (Project Manager)	All Team Members	Weekly	E-mail, etc	Every Monday
Project Effort Reports	All Teams Members	All Team Members	Weekly	E-mail,whatsapp etc	Every Friday
Milestone Review Meetings	Vineeth Reddy Kodakandla (Project Manager)	All Team Members, Course Instructor	Per Milestone	Face-to-face	As per milestone schedule
Code Review Sessions	Sricharitha Katta (Technical Manager)	All Team Members	Weekly	GitHub, Zoom	Every Wednesday

7. Project Members and Roles

The following table identifies the project team members, their assigned roles, and their respective responsibilities within the PathWise AI project.

Project Role	Name	Responsibilities
Project Manager	Vineeth Reddy Kodakandla	Oversees the overall project execution, manages the project schedule and milestones, coordinates team communication, ensures deliverables meet quality standards, manages project risks, and serves as the primary liaison with the course instructor.
Project Technical Manager	Sricharitha Katta	Manages the technical architecture and technology stack decisions, oversees the microservices design, coordinates SDN controller integration, manages the CI/CD pipeline and Docker containerization, and ensures technical consistency across all system components.
Project Requirements Manager	Meghana Nalluri	Leads the requirements elicitation and analysis process, documents stakeholder needs, manages the Software Requirements Specification (SRS), ensures traceability of requirements throughout the project lifecycle, and validates that the implemented system meets stakeholder expectations.
Project Design Manager	Bharadwaj Jakkula	Leads the system architecture and software design activities, creates the Software Design Document (SDD), designs the LSTM model architecture and data pipeline, designs the frontend dashboard and user interface, and ensures design consistency across all modules.
Project Test Manager	Bharadwaj Jakkula	Develops the test strategy and test plan, designs test cases for all system components, manages the Software Test Document (STD), coordinates integration and system testing, and ensures that all quality goals and acceptance criteria are met.

Project Configuration Manager	Sricharitha Katta	Manages the version control system (Git/GitHub), establishes branching and merging strategies, maintains configuration baselines, manages build automation, and ensures all project artifacts are properly versioned and stored.

8. Project Work Breakdown Structure (WBS)

The following table presents the work breakdown structure for the PathWise AI project. Each work package represents a group of related tasks with an assigned responsible team member and estimated effort.

Work Package ID	Work Package Name	Responsible Team Member	Estimated Effort (Man*Hour)	Definition
WP-1	Project Management	Vineeth Reddy Kodakandla	80	Includes project planning, scheduling, progress tracking, milestone reviews, risk management, stakeholder communication, and team coordination activities throughout the project lifecycle.
WP-2	Requirements Engineering	Meghana Nalluri	80	Encompasses requirements elicitation from stakeholders, requirements analysis and modeling, creation of the Software Requirements Specification (SRS), requirements validation, and traceability matrix development.
WP-3	System Architecture and Design	Bharadwaj Jakkula	90	Covers the overall system architecture design, microservices decomposition, database schema design, API specification, LSTM model architecture, frontend UI/UX design, and creation of the Software Design Document (SDD).
WP-4	Predictive Telemetry Engine Development	Sricharitha Katta	60	Involves development of the LSTM neural network model, telemetry data ingestion pipeline, model training and validation, predictive health score generation, and integration with TimescaleDB for time-series data storage.
WP-5	Autonomous Traffic Steering Development	Vineeth Reddy Kodakandla	40	Covers SDN controller integration with OpenDaylight/ONOS, flow table modification logic, hitless handoff implementation, session state management, and real-time traffic rerouting based on AI predictions.
WP-6	Digital Twin Sandbox Development	Sricharitha Katta	40	Includes Mininet network topology mirroring, Batfish policy compliance validation, loop

				detection algorithms, sandbox simulation execution, and audit trail generation for validated routing changes.
WP-7	Intent-Based Management Interface Development	Bharadwaj Jakkula	30	Encompasses the natural language processing interface, YANG/NETCONF configuration translation, React-based dashboard development, D3.js visualization components, and user interaction design.
WP-8	Multi-Link Health Scoreboard Development	Meghana Nalluri	30	Covers real-time health score visualization, AI confidence level display, automated path switch reasoning presentation, D3.js interactive charts, and dashboard integration with the backend API.
WP-9	Backend API and Integration Layer	Vineeth Reddy Kodakandla	40	Involves FastAPI service development, Redis pub/sub messaging integration, REST API endpoints, authentication and authorization, and integration with SDN controllers and telemetry sources.
WP-10	Testing and Quality Assurance	Bharadwaj Jakkula	40	Encompasses unit testing, integration testing, system testing, LSTM model accuracy validation, performance testing, creation of the Software Test Document (STD), and defect tracking and resolution.
WP-11	Configuration Management and DevOps	Sricharitha Katta	20	Covers Git repository management, branching strategy enforcement, Docker containerization, CI/CD pipeline configuration, build automation, and artifact version control.
WP-12	Documentation and Presentations	All Team Members	50	Includes preparation of all project documents (PVD, PP, SRS, SDD, STD), final project report compilation, and project presentation preparation and delivery.

9. Project Deliverables

Deliverable Number	Deliverable Name
D-1	Team Project Proposal
D-2	Project Vision Document (PVD)
D-3	Project Plan (PP)

D-4	Software Requirements Specification (SRS)
D-5	Software Design Document (SDD)
D-6	Predictive Telemetry Engine (LSTM Model and Data Pipeline)
D-7	Autonomous Traffic Steering Module (SDN Integration)
D-8	Digital Twin Validation Sandbox (Mininet/Batfish)
D-9	Intent-Based Management Interface (React Dashboard with NLP)
D-10	Multi-Link Health Scoreboard (Real-Time Visualization)
D-11	Backend API and Integration Layer (FastAPI Services)
D-12	Software Test Document (STD)
D-13	Complete Source Code Repository
D-14	Docker Containerization and Deployment Configuration
D-15	Final Project Presentation

10. Project Risks

The following table identifies the key risks associated with the PathWise AI project, stated in the if–then format, along with the corresponding risk reduction approaches.

Risk Number	Risk	Risk Reduction Approach
R-1	If the LSTM model cannot achieve the target 90% prediction accuracy due to insufficient or noisy training data, then the Predictive Telemetry Engine will produce unreliable forecasts, undermining the core value proposition of proactive traffic steering.	The team will use publicly available network telemetry datasets for initial training and supplement with synthetic data generated through Mininet simulations. The model architecture will be iteratively refined using hyperparameter tuning and cross-validation to maximize accuracy within the available data constraints.
R-2	If the integration with SDN controllers (OpenDaylight or ONOS) encounters API compatibility issues or version conflicts, then the Autonomous Traffic Steering feature will be unable to execute hitless handoffs in the production environment.	The team will begin SDN controller integration early in the development cycle and maintain compatibility with both OpenDaylight and ONOS to provide fallback options. Comprehensive integration tests will be developed and executed regularly to detect compatibility issues early.
R-3	If the Digital Twin Validation Sandbox using Mininet and Batfish cannot accurately mirror complex real-world network topologies, then the safety validation layer will provide false assurance and AI-driven changes may still destabilize the network.	The team will validate the sandbox against known network configurations with predictable behavior. Test scenarios will progress from simple topologies to more complex multi-site configurations, and validation accuracy will be measured against expected outcomes.
R-4	If team members face scheduling conflicts due to academic workload or other commitments, then project milestones may be delayed and the quality of deliverables may be compromised.	The project manager will maintain a detailed schedule with built-in buffer time for each milestone. Weekly progress meetings will identify potential delays early. Tasks will be distributed to ensure no single team member is

		a critical bottleneck, and cross-training will be provided for key modules.
R-5	If the natural language processing component of the Intent-Based Networking interface cannot reliably translate user commands into accurate YANG/NETCONF configurations, then generalist IT staff will be unable to effectively manage network policies.	The team will define a controlled vocabulary of supported natural language patterns and implement template-based translation for common commands. User testing with non-specialist participants will be conducted to validate usability and accuracy of the translation engine.
R-6	If the real-time telemetry ingestion pipeline introduces latency that exceeds the 50-millisecond failover target, then the system will be unable to achieve hitless handoffs and traffic disruption will occur during link transitions.	The team will use Redis for low-latency pub/sub messaging and TimescaleDB optimized for time-series writes. Performance benchmarks will be established early and the pipeline will be load-tested to ensure sub-50ms end-to-end processing under expected traffic volumes.
R-7	If the project scope exceeds the available 600 man-hours budget, then lower-priority features such as the Intent-Based Management Interface may not be fully implemented within the academic semester timeline.	The team will follow the feature priority defined in the Project Vision Document, focusing first on the three high-priority features (Predictive Telemetry Engine, Autonomous Traffic Steering, and Digital Twin Validation Sandbox). Lower-priority features will be implemented incrementally, with a minimum viable version delivered if time constraints arise.